

# **MDS-UPDRS Finger Tapping Score**

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

**Instructions to the examiner:**

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

**Scoring Criteria:**

**0 (Normal):** No problems

**1 (Slight):** Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

**2 (Mild):** Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

**3 (Moderate):** Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

**4 (Severe):** Cannot or can only barely perform the task due to slowing, interruptions, or decrements

**Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.**

## **## EXTRACTED FEATURES**

**Number of taps: 11**

**Tap periods (frames): [123, 110, 97, 104, 92, 118, 101, 97, 92, 99]**

**Tap amplitudes (pixels): [129.65338406690356, 129.5414991421668, 115.83177456984762, 110.01818031580054, 116.84605256490268, 107.33592129385204, 95.01578816175763, 116.76472069936193, 116.8289347721702, 114.12712210513327, 116.81181447096864]**

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**GPT-5**

## **Analysis of extracted features for MDS-UPDRS 3.4 (Finger Tapping)**

Given task: 10 taps as quickly and as large as possible.

Provided data:

- Number of taps detected: 11 (suggests one extra segmentation, but we evaluate the 10-tap sequence per instruction).
- Tap periods (frames) for the 10 intervals between taps: [123, 110, 97, 104, 92, 118, 101, 97, 92, 99]
- Tap amplitudes (pixels) per tap (11 values): [129.65, 129.54, 115.83, 110.02, 116.85, 107.34, 95.02, 116.76, 116.83, 114.13, 116.81]

Interpretation approach:

- Speed: assess slowing via average period and variability; identify interruptions/hesitations via conspicuous prolongations compared to neighbors.
- Rhythm interruptions: count evident hesitations (notable longer periods).
- Amplitude trend: evaluate whether amplitude decrement occurs and when it begins (near end, mid, or after 1st tap).

### **1) Speed and rhythm/hesitations**

- Periods range: min 92, max 123 frames. Relative variability is modest, but there are a few longer intervals (123, 118, 110) interspersed with faster ones (92–104).
- Candidate hesitations: 123 (start), 118 (mid-late), and 110 (early). These are not extreme outliers but are clear local lengthenings relative to adjacent periods.
- Count of interruptions/hesitations: approximately 3 discernible longer intervals across the 10 periods. No evidence of a longer arrest/freezing.

This pattern aligns with “3 to 5 interruptions” rather than only “one or two,” supporting a score of at least 2 (Mild) on the interruptions criterion.

## 2) Amplitude behavior (decrement timing)

- Amplitudes per tap show initial high values for taps 1–2 (~129.6), then a drop by tap 3–4 (115.8 → 110.0), further reduction around tap 6–7 (107.3 → 95.0), and then partial recovery (~116–117) for taps 8–11.
- This indicates a decrement beginning early (by tap 3–4), i.e., well before the end and even before the midpoint, though there’s partial recovery later.
- According to criteria:
  - “Decrements near the end” → score 1
  - “Decrements midway through” → score 2
  - “Decrements starting after the 1st tap” → score 3Given the clear drop from tap 2 to taps 3–4, the decrement starts after the 1st tap, supporting score 3 on the amplitude criterion.

## 3) Slowing

- Periods are not uniformly prolonged and there’s no strong indication of moderate or severe slowing; it appears closer to slight-to-mild. This alone wouldn’t raise the score beyond 2, but scoring is determined by the most severe applicable criterion among interruptions, slowing, and decrement.

Overall determination:

- Interruptions: roughly 3 hesitations → consistent with score 2.
- Amplitude: decrement starts after the 1st tap → consistent with score 3.
- No long arrest/freezing; task clearly performed → not score 4.

Final score is driven by the highest applicable criterion.

Score: 3

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